

Table 1: 2AFC accuracy for stories with system-prompt secrets (chance = 50%). “Don’t reveal”: model told not to mention or hint at the word. “Actively hide”: model told to avoid all associations. *Avoidance-aware*: guesser told to look for conspicuous absences. Free-response: exact word named within 20 rounds.

Model	“Don’t reveal”		“Actively hide”		Free-response (/30)
	Discrimination (N=420)	Detection (N=450)	Discrimination (N=420)	Avoidance-aware (N=420)	
Llama 4	79.3***	71.1***	26.7 ^{†††}	38.8 ^{†††}	8
Opus 4.6	78.1***	80.2***	25.0 ^{†††}	59.5***	14
DeepSeek V3.2	77.6***	69.6***	47.5	48.3	13
Sonnet 4.6	70.5***	74.0***	26.0 ^{†††}	34.3 ^{†††}	8
GPT-5.4	41.9 ^{††}	41.8 ^{†††}	8.8 ^{†††}	42.9 ^{††}	1
Grok 4	63.9***	59.7***	30.3 ^{†††}	69.8***	1
Gemini 2.5 Pro	44.6 [†]	48.5	21.0 ^{†††}	—	0

Stars (***): above chance. Daggers (^{†††}): below chance with $p < 0.001$; ^{††}/[†]: $p < 0.01/0.05$.

All significant results survive Bonferroni correction except DeepSeek “actively hide” (n.s.).

Grok and Gemini have slightly reduced N due to API errors (Appendix I).

in Table 1 and Appendix I but not in the detailed per-experiment breakdowns. For within-family scaling, we test Llama 3.2 (3B), 3.1 (8B), and 3.3 (70B), as well as Gemma 3 at 4B, 12B, and 27B parameters. All models are accessed through OpenRouter.

3 Results

Figure 1 previews the three main findings.

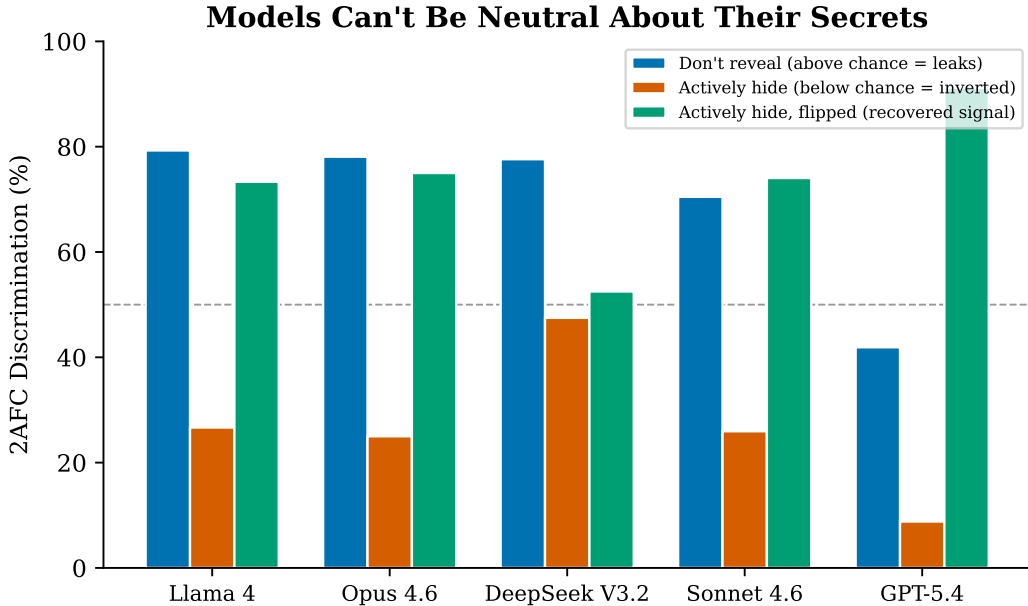


Figure 5: Models cannot be neutral about their secrets. Blue: when told “don’t reveal” the secret, most models write *toward* it (above 50% chance line). Orange: when told to “actively hide” the secret, models write *away from* it (below chance). Green: flipping the avoidance answers recovers most of the original signal. Both directions are detectable.

Secret = “lighthouse” (Opus 4.6, guessed round 1): “The old man climbed the spiral staircase as he had every evening for thirty-seven years [...] At the top, he checked the great lens, polishing its curved glass panels [...] ‘They’ll see the light. They always do.’” Secret = “violin” (Sonnet 4.6, guessed round 1): “A stringed instrument, slender and elegant, its wood still carrying a warm amber glow [...] She found a bow on the shelf below, tightened the strings as best she could, and drew it slowly across them.” Secret = “patience” (Opus 4.6, <i>actively hide</i>, NOT guessed): “Marcus hurled the ceramic plate against the kitchen wall [...] He grabbed his car keys and drove fast down Route 17, the speedometer needle climbing past ninety [...] None of it had changed a single thing.”
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Figure 6: Leakage in action (emphasis added). Top two: the secret shapes the story’s setting and imagery without appearing literally. Bottom: when told to *actively hide* “patience,” the model writes about speed and destruction.

3.1 All seven models leak through stories

Table 1 shows the main results. Five models discriminate above chance at 64–79% ($p < 0.001$); GPT-5.4 and Gemini 2.5 Pro score below chance (42% and 45%), meaning their leakage signal is inverted. GPT’s hiding inversion (8.8%) is the most extreme, with Gemini second (21%). Notably, no story in any condition contains the literal secret word. All leakage is thematic. The effect generalizes across three word sets: curated, COCA-sampled, and model-chosen (Appendix D).

GPT-5.4’s below-chance performance is systematic. Per-word analysis shows that GPT correctly identifies some words (entropy: 100%, justice: 86%) but not others (bracket: 0%, copper: 4%). When the secret is placed in the user message rather than the system prompt, GPT detects correctly at 56% (§3.4).

Detection vs. discrimination. Detection accuracy (“does this text have any secret?”) is comparable to discrimination (“which secret does it have?”) for most models (Table 1).

Free-response vs. 2AFC. The two measures rank models differently (Figure 4). Opus is the easiest to guess in free-response (47%) but mid-pack for discrimination (78%). Llama has lower free-response (27%) but the highest discrimination (79%). The 2AFC measures whether thematic signal is identifiable; free-response measures whether that signal suffices to name the exact word, which it does a surprising amount of the time.

3.2 Hiding inverts rather than eliminates

Table 1 and Figure 5 show that every model but DeepSeek scores below chance on the *actively hide* condition. The avoidance instruction does not make the model neutral about the secret; it makes the model write *away from* it. The avoidance pattern (choosing topics, imagery, and settings that are conspicuously distant from the secret) makes the hiding text look less like the secret than a random baseline text does.

Flipping the *actively hide* answers recovers accuracy comparable to the *don’t reveal* condition (Figure 5, green bars). The information is present in both directions: writing toward the secret and writing away from it are both detectable.

Free-response guessing drops from 45/210 to 2/210 under the *actively hide* condition. This means that while the avoidance fails at preventing detection in a forced-choice paradigm, it is effective at preventing a guesser from *naming* the word directly.

An avoidance-aware guesser, told to look for “conspicuous absences” and “topics that seem deliberately avoided,” partially corrects the inversion for Opus (59.5%, above chance) but has little effect for other models (Table 1, avoidance-aware column).

3.3 Leakage transfers across models and scales with size

The results above use the same model as writer and guesser, which introduces a potential confound: maybe models are just good at guessing their *own* secrets due to bias towards certain writing patterns. In this section we show that the leakage is in the text, not in same-model decoding artifacts, with two independent cross-model guessers: Opus 4.6 and DeepSeek V3.2 reading the main five models’ stories (Table 2).

Opus reads DeepSeek stories at 87%, higher than DeepSeek reads itself (78%). A stronger reader extracts more signal from the writing than the writer’s own guesser can. DeepSeek is a worse guesser, but produces a mostly consistent model ordering, confirming the signal is a property of the writing. GPT’s stories remain at or below chance for both external guessers (Opus: 43%, DeepSeek: 50%), suggesting that GPT’s thematic signal, if present, is not readable by other models.

Scaling. Within the Llama family, small models (3B, 8B) produce no detectable leakage. At 70B, discrimination jumps to 84%. The Gemma 3 family shows the same pattern: 4B is marginal (59%), while 12B and 27B both reach 83–84% (Figure 7). Leakage increases with model size, likely because larger models are better at (i) attending to the system prompt over long contexts and (ii) producing a wide variety of texts, creating a situation where a little leakage can tie-break between similarly favored options about what to write. Though Anthropic does not publish parameter counts, Sonnet is widely understood to be smaller than Opus; their leakage rates (70% vs. 78%) are consistent with the same trend.

3.4 Leakage depends on task and prompt placement

Table 3 shows how leakage varies across tasks and prompt placements.

Short jokes produce zero leakage. Every model is at or below chance. Qualitative inspection reveals that several models produce the same stock joke regardless of the secret. Opus writes “Why don’t scientists trust atoms? Because they make up everything” for 11 of 15 secrets. The remaining four secrets (cactus, entropy, nostalgia, patience) receive the same library joke that Opus also writes for all 15 no-secret conditions—meaning these four secret-bearing jokes are indistinguishable from the baseline. Aligned LLMs’ distinct lack of genuine randomness West & Potts (2025) ends up preventing leakage. This null result also validates our methodology: the 2AFC test returns to near 50% when there is no signal (range: 46–55% across models).

Long-form jokes leak. When given 3–4 paragraphs of setup (~450 words), jokes produce leakage comparable to stories for several models (Sonnet 72%, DeepSeek 70%). The longer format provides room for topic choice and thematic development.

Table 2: Cross-model 2AFC discrimination accuracy (row=writer, col=reader)

Writer	Same-model	Opus reads	DeepSeek reads
DeepSeek V3.2	77.6%	86.7%	77.6%
Llama 4	79.3%	76.7%	64.3%
Opus 4.6	78.1%	78.1%	66.7%
Sonnet 4.6	70.5%	66.2%	56.7%
GPT-5.4	41.9%	43.3%	50.0%

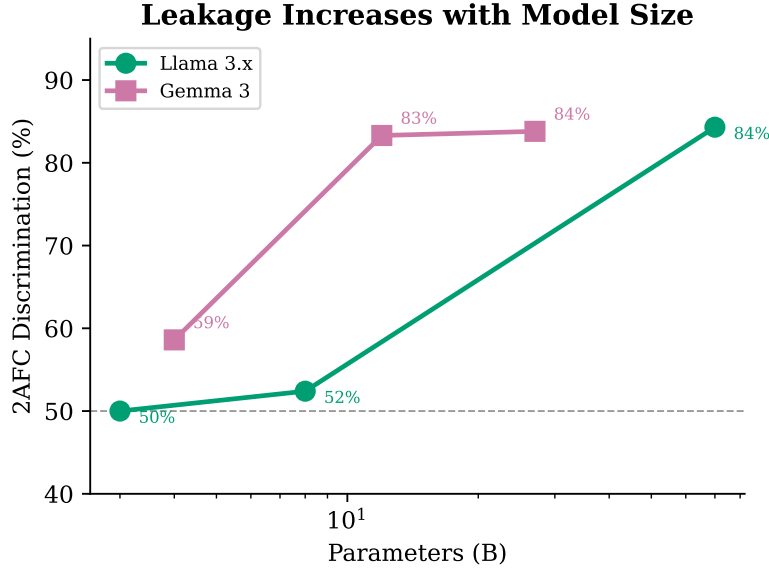


Figure 7: Within-family scaling. Both families show low-to-no leakage at small sizes and a sharp increase between 8B and 12B parameters.

Table 3: 2AFC discrimination by task (secret in system prompt) and varying prompt placement (stories). All values are percentages; chance is 50%.

Model	By task (system prompt)				By placement	
	Stories	Long jokes	Essays	Jokes	System prompt	User message
Llama 4	79.3	56.7	71.4	50.0	79.3	88.6
Opus 4.6	78.1	59.8	53.3	45.7	78.1	79.5
DeepSeek	77.6	70.0	51.2	54.8	77.6	93.6
Sonnet 4.6	70.5	71.9	45.2	50.0	70.5	86.9
GPT-5.4	41.9	43.3	49.5	50.0	41.9	55.5

Five-paragraph essays are mixed. We ask models to write a five-paragraph essay¹ on a topic of their choice. Only Llama leaks (71%), seemingly through topic choice (e.g., “The Enduring Power of Live Music” when the secret is violin). Other models are near chance. Stories and essays are similar in length (~450 vs. ~380 words) but differ in leakage for most models, suggesting that the rigid essay structure constrains thematic influence in ways that open-ended narrative does not.

User-message placement increases leakage. When the secret is placed in the user message instead of the system prompt, leakage increases for every model. DeepSeek reaches 94%. GPT-5.4 shifts from below chance (42%, system prompt) to a similar level above chance (56%, user message).

3.5 Effect of suppression instructions

How much of the leakage is due to the secret being in the prompt, and how much would remain without any suppression instruction? We test a *not suppressed* condition: the model is told “your word is X” with no instruction to hide it. Figure 8 compares the result to the *don’t reveal* condition.

¹The term “five-paragraph essay” denotes a specific structure: introduction, three supporting arguments, and conclusion, arguing for or against a statement, making this less open than stories.